How to Build a Time Series Agent

Reuse what is general · adapt what is temporal · rebuild what is task-bound



Scenario & data

series · context · task request

LLM



ALIGN

interprets temporal evidence and selects actions

- Represent numbers for the language model
- Reason over temporal evidence
- Learn which workflow action to take

Not enough alone: it acts through the harness

General Harness



REUSE

domain-independent runtime

- Interaction & memory
- Tools & execution
- Planning & coordination
- Feedback & self-improvement

Carry it over from general Agents

General Time Series Harness



ADAPT

temporal semantics shared across tasks

- Timestamps, units, missingness
- Windows, resampling, decomposition
- Only information available at t_0
- Chronology guards & validators

Add time where a generic runtime is ambiguous

Task-Specific Harness



REBUILD

procedures bound to one task contract

- Forecasting
- Synthesis & augmentation
- Anomaly diagnosis
- Decision support

Rebuild around output, objective and actions

Evaluation



VERIFY

tests the complete workflow

- Process validity
- Temporal validity
- Task performance
- Decision utility

Valid run ≠ accurate result ≠ useful decision



Accepted output

forecast · diagnosis · decision

↺ feedback from validation: revise the plan, or terminate



reuse, adapt, rebuild, verify

Key claim: Align model capability, harness responsibility and evaluation